WORKED SOLUTION

STEP II 1991, Question 11

Mechanics • Projectile motion

STEP II 1991, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/91-s2-q11

The key idea

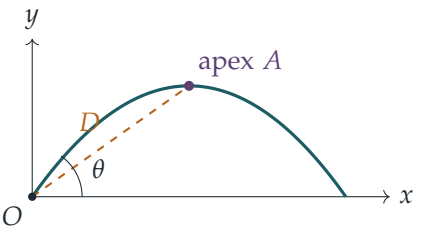
Work with the *square* of the distance from the gun: it avoids square roots and turns both parts into polynomial problems. In the first part the apex distance squared is a quadratic in $\sin^2\theta$, so completing the square (or differentiating) finds the best angle. In the second part, the shell gets nearer exactly when the distance squared is decreasing, so we look at the sign of its time derivative along the whole flight.

Solution

The model

We treat the shell as a particle moving freely under gravity, with no air resistance, fired from the origin O (where the gunners stand) with speed u at elevation θ, where $0 < \theta < \frac{\pi}{2}$. Take x horizontal and y vertically up. At time t,

$$x = ut\cos\theta, \qquad y = ut\sin\theta - \tfrac{1}{2}gt^2.$$



The angle of elevation

The apex is reached when $\dot{y} = u\sin\theta - gt = 0$, that is at $t = u\sin\theta/g$. There

$$x_A = \frac{u^2\sin\theta\cos\theta}{g}, \qquad y_A = \frac{u^2\sin^2\theta}{2g}.$$

Write D for the distance OA and put $s = \sin^2\theta$, so that $\cos^2\theta = 1 - s$. Then

$$\begin{aligned} D^2 = x_A^2 + y_A^2 &= \frac{u^4}{g^2}\left(\sin^2\theta\cos^2\theta + \tfrac{1}{4}\sin^4\theta\right) \\ &= \frac{u^4}{g^2}\left(s(1-s) + \tfrac{1}{4}s^2\right) = \frac{u^4}{g^2}\left(s - \tfrac{3}{4}s^2\right). \end{aligned}$$

Since the damage decreases with distance, the gunners want D as large as possible, which is the same as making D^2 as large as possible. Completing the square,

$$s - \tfrac{3}{4}s^2 = \tfrac{1}{3} - \tfrac{3}{4}\left(s - \tfrac{2}{3}\right)^2,$$

which is largest, with value $\frac{1}{3}$, exactly when $s = \frac{2}{3}$. This value is allowed, since $s = \sin^2\theta$ takes every value in $(0, 1)$. So $\sin^2\theta = \frac{2}{3}$ and $\cos^2\theta = \frac{1}{3}$, giving

$$\boxed{\tan\theta = \sqrt{2}, \qquad \theta = \arctan\sqrt{2} \approx 54.7^\circ}$$

and then the apex is at distance $D = \dfrac{u^2}{\sqrt{3}\,g}$ from the gun.

Does the shell come nearer during its descent?

Let $r^2 = x^2 + y^2$ be the squared distance from the gun at time t. The shell gets nearer exactly when r^2 is decreasing, so we need the sign of

$$\tfrac{1}{2}\frac{\mathrm{d}}{\mathrm{d}t}(r^2) = x\dot{x} + y\dot{y}.$$

Ask yourself Why not just compare the distance at the apex with the distance at a few later points?

A few points can't rule out a dip in between. The derivative tells us about *every* instant of the flight at once. With $\dot{x} = u\cos\theta$ and $\dot{y} = u\sin\theta - gt$,

$$\begin{aligned} x\dot{x} + y\dot{y} &= u^2 t\cos^2\theta + \left(ut\sin\theta - \tfrac{1}{2}gt^2\right)(u\sin\theta - gt) \\ &= u^2 t - \tfrac{3}{2}ugt^2\sin\theta + \tfrac{1}{2}g^2t^3 \\ &= \tfrac{1}{2}t\left[(gt)^2 - 3u\sin\theta\,(gt) + 2u^2\right], \end{aligned}$$

using $\cos^2\theta + \sin^2\theta = 1$. For $t > 0$ the sign is that of the bracket, a quadratic in gt. Completing the square,

$$(gt)^2 - 3u\sin\theta\,(gt) + 2u^2 = \left(gt - \tfrac{3}{2}u\sin\theta\right)^2 + u^2\left(2 - \tfrac{9}{4}\sin^2\theta\right).$$

At the chosen angle $\sin^2\theta = \frac{2}{3}$, so the constant term is $u^2\left(2 - \frac{3}{2}\right) = \frac{1}{2}u^2 > 0$. The bracket is therefore strictly positive for every t, so r^2 is strictly increasing throughout the flight.

So **no**: the shell moves steadily further from the gunners all the way down, and the nearest point of the descending path is the apex itself. This shows that the gunners' shortcut of considering only an explosion at the maximum height was justified: for this angle it really does give the nearest point of the descent, at distance $u^2/(\sqrt{3}\,g)$.

Checks

- For $\theta = \frac{\pi}{4}$ the apex is at $(u^2/2g,\, u^2/4g)$, so $D^2 = \frac{5}{16}u^4/g^2$, a little less than our maximum $\frac{1}{3}u^4/g^2$, as it should be.
- The general calculation shows that the shell first comes nearer only when $\frac{9}{4}\sin^2\theta \geq 2$, i.e. $\sin\theta \geq \frac{2\sqrt{2}}{3}$, an angle of about $70.5°$, comfortably above $54.7°$. (The limit $\theta \to \frac{\pi}{2}$ makes sense: a shell fired straight up falls straight back towards the gun.)
- Numerically, with $u = g = 1$, a search over θ gives the largest apex distance $0.57735 \approx 1/\sqrt{3}$ at $\theta \approx 54.736°$, and stepping along the trajectory at that angle shows the distance from O never decreasing; at $70.5° \pm 0.6°$ it switches from monotonic to not.

Common slips

- Maximising the horizontal distance or the height of the apex instead of its straight-line distance from the gun.
- Assuming the distance from the gun obviously increases after the apex because the shell keeps moving away horizontally. It is falling at the same time, and for steep enough shots it really does come back nearer.

Beyond STEP

The trick of differentiating r^2 rather than r, and reading off the sign of $\mathbf{r}\cdot\dot{\mathbf{r}}$, is the standard way to decide whether a moving point is approaching or receding, and it appears throughout vector mechanics and in the study of orbits. The threshold angle $\arcsin(2\sqrt{2}/3)$ is a classic result: below it, a projectile's distance from the launch point always increases. Kleppner and Kolenkow, *An Introduction to Mechanics*, develops the vector approach to motion that makes this kind of argument natural.

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